

From Causal Discovery to Dynamic Causal Inference in Neural Time Series

Dmitry Zaytsev
Lucy Family Institute for Data &
Society
University of Notre Dame
Notre Dame, Indiana, USA
zaytsevd2@gmail.com

Valentina V. Kuskova
Lucy Family Institute for Data &
Society
University of Notre Dame
Notre Dame, Indiana, USA
vkuskova@nd.edu

Michael Coppedge
Political Science
University of Notre Dame
Notre Dame, Indiana, USA
mcoppedg@nd.edu

Abstract

Time-varying causal models provide a powerful framework for studying dynamic scientific systems, yet most existing approaches assume that the underlying causal network is known a priori - an assumption rarely satisfied in real-world domains where causal structure is uncertain, evolving, or only indirectly observable. This limits the applicability of dynamic causal inference in many scientific settings. We propose Dynamic Causal Network Autoregression (DCNAR), a two-stage neural causal modeling framework that integrates data-driven causal discovery with time-varying causal inference. In the first stage, a neural autoregressive causal discovery model learns a sparse directed causal network from multivariate time series. In the second stage, this learned structure is used as a structural prior for a time-varying neural network autoregression, enabling dynamic estimation of causal influence without requiring pre-specified network structure. We evaluate the scientific validity of DCNAR using behavioral diagnostics that assess causal necessity, temporal stability, and sensitivity to structural change, rather than predictive accuracy alone. Experiments on multi-country panel time-series data demonstrate that learned causal networks yield more stable and behaviorally meaningful dynamic causal inferences than coefficient-based or structure-free alternatives, even when forecasting performance is comparable. These results position DCNAR as a general framework for using AI as a scientific instrument for dynamic causal reasoning under structural uncertainty.

CCS Concepts

• **Computing methodologies** → **Causal reasoning and diagnostics; Time series analysis; Machine learning.**

Keywords

Dynamic causal inference, Time-varying autoregression, Impulse response analysis, Counterfactual simulation, Structural priors

ACM Reference Format:

Dmitry Zaytsev, Valentina V. Kuskova, and Michael Coppedge. 2026. From Causal Discovery to Dynamic Causal Inference in Neural Time Series. In *Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining V.2 (KDD '26)*, August 09–13, 2026, Jeju Island, Republic of Korea. ACM, New York, NY, USA, 11 pages. <https://doi.org/10.1145/3770855.3818956>



This work is licensed under a Creative Commons Attribution 4.0 International License. *KDD '26, Jeju Island, Republic of Korea*
© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2259-2/2026/08
<https://doi.org/10.1145/3770855.3818956>

1 Introduction

Scientific models are not evaluated solely by how accurately they predict future observations, but by what they reveal about the underlying processes that generate data. This distinction is especially important in the social sciences and economics, where researchers seek to model human systems as complex adaptive processes and to evaluate the dynamic consequences of policy-relevant interventions through counterfactual simulation rather than prediction alone. In such settings, predictive accuracy is a necessary condition for model credibility, but it is not sufficient for scientific explanation.

Dynamic causal understanding therefore relies on forms of model interrogation that go beyond forecast performance [10]. Scientists examine impulse response functions [47, 55] to study how shocks propagate over time, counterfactual trajectories [51, 72] to assess the effects of alternative interventions, and the stability and persistence of causal effects across regimes [13]. These tools support explanation, hypothesis testing, and early warning in ways that aggregate error metrics cannot capture.

Despite this, much of modern machine learning for time series prioritizes predictive performance [48]. Flexible neural models can achieve strong forecasting results [24] but often behave poorly under causal interrogation [42]. Impulse responses may exhibit oscillations or sign reversals [39, 53], counterfactual trajectories may diverge unrealistically [70], and inferred dynamics can be highly sensitive to small perturbations [40]. Consequently, such models are difficult to use as scientific instruments, even when their predictive accuracy is competitive.

Existing frameworks for dynamic causal inference face a complementary limitation. Models that provide interpretable time-varying causal parameters, such as time-varying vector autoregressions or network autoregressive models, typically assume that the underlying causal structure is known in advance [21]. In many domains, including political systems [7], ecological networks [19], and social processes [58], this assumption is rarely satisfied. Causal structure is often uncertain, evolving, or itself the object of investigation. Researchers are therefore forced to choose between interpretable dynamic models with questionable structural assumptions and flexible predictive models that offer little causal insight [23, 61, 65].

This paper introduces *Dynamic Causal Network Autoregression* (DCNAR), a framework designed to bridge this divide. DCNAR enables dynamic causal analysis when causal structure is unknown by integrating data-driven causal discovery with time-varying network autoregression in a principled pipeline. Rather than optimizing exclusively for predictive accuracy, DCNAR is designed to support scientifically meaningful interrogation through impulse responses

and counterfactual analysis, accepting modest trade-offs in predictive optimality in exchange for greater interpretability, stability, and theoretical coherence of inferred causal dynamics.

Our methodology is motivated by domain-specific democracy panel data characterized by many short, heterogeneous country-level time series, whose constraints directly inform the design of DCNAR and its emphasis on structural regularization. Using multi-country panel time-series data as a motivating scientific application, we show that DCNAR produces impulse responses and counterfactual trajectories that are stable, interpretable, and consistent with theoretical expectations, even when causal structure is not known a priori. While DCNAR does not dominate all baselines on every predictive metric, it remains competitively calibrated and avoids the pathological causal behavior often exhibited by purely predictive models. These properties make DCNAR suitable not merely as a forecasting tool, but as a methodological instrument for scientific reasoning about dynamic systems.

The contributions of this work are twofold. First, we demonstrate that DCNAR achieves competitive predictive and distributional performance relative to common time-series baselines while maintaining probabilistic calibration. Second, and more importantly, we show that DCNAR uniquely supports interpretable, stable, and theoretically meaningful impulse-response and counterfactual analysis in settings with unknown causal structure, enabling forms of dynamic causal understanding that are inaccessible to existing approaches.

2 Scientific Challenge: Dynamic Causality Without Known Structure

Many scientific domains seek to understand not only whether variables are related, but how causal relationships evolve over time [68]. In complex systems [71] such as political regimes or social processes, causal influence is rarely static: interactions may strengthen, weaken, or reverse as contextual conditions change [2]. Capturing such temporal variation is therefore essential for scientific explanation, early warning, and counterfactual reasoning [23, 36]. Time-varying causal models, including time-varying vector autoregression [33] and network autoregression [3, 75], have been widely developed for this purpose [14].

Despite their conceptual appeal, the applicability of dynamic causal models is constrained by a central assumption: the underlying causal network is known in advance [20, 21, 60]. Most formulations require researchers to specify which directed interactions are admissible based on theory, expert knowledge, or prior empirical studies. In many scientific domains, however, causal structure is precisely what is uncertain, contested, or under investigation [28, 43]. As a result, researchers face a persistent trade-off between imposing potentially incorrect structural assumptions or abandoning explicit causal modeling in favor of flexible predictive approaches that sacrifice interpretability and hypothesis testing [1, 18].

At a deeper level, this trade-off reflects a structural rather than algorithmic limitation. Joint estimation of time-varying causal effects and network structure is ill-posed in finite, high-dimensional time series, while unconstrained dynamic models are unstable and difficult to interpret. Existing approaches therefore stabilize estimation by fixing structure [8], typically treating causal discovery and

dynamic inference as separate problems [30]. This separation transforms dynamic causal models into descriptive tools conditioned on assumed networks [27], limiting their use for evaluating or discovering causal hypotheses. The challenge addressed in this work is to overcome this limitation by treating causal structure not as a fixed prerequisite, but as a learned and empirically testable component of dynamic causal inference.

3 Scientific Evidence in Dynamic Causal Models

Dynamic causal models are often evaluated using predictive criteria such as forecast error or likelihood [59], but these metrics alone are insufficient for scientific inference. In practice, researchers assess causal models by how they behave under perturbation, examining impulse responses, counterfactual trajectories [5, 34], function-valued influence [45], and the stability of inferred relationships across contexts. Such behavioral probes reveal how effects propagate through a system over time and provide forms of evidence essential for explanation and hypothesis evaluation that cannot be reduced to predictive performance.

3.1 Limits of prediction-centric evaluation

Prediction-focused evaluation treats the data-generating process as a black box [49], emphasizing forecast accuracy [52] without regard to the mechanisms through which predictions are achieved. For causal analysis, this criterion is too weak [12]. Models with similar predictive accuracy may encode fundamentally different assumptions about feedback, temporal dependence, and causal pathways, leading to divergent conclusions when used for explanation or intervention analysis [18, 35].

These limitations are especially pronounced for flexible machine learning models. Highly expressive architectures can interpolate observed data effectively [64], yet exhibit unstable or implausible behavior under causal interrogation [57]. Small perturbations may induce oscillatory impulse responses, sign reversals, or explosive counterfactual trajectories that are difficult to reconcile with domain knowledge [62, 67]. Consequently, predictive accuracy should be viewed as a necessary but not sufficient condition for scientific usefulness in dynamic causal modeling. Additional criteria are required to assess whether inferred dynamics are coherent, interpretable, and robust.

Moreover, most evaluations of such models have been conducted in technical or engineered domains with controlled dynamics and abundant data [67]. In the social sciences, where systems are adaptive, data are observational, and causal structure is uncertain, systematic evidence on how flexible predictive models behave under causal interrogation remains limited [62].

3.2 Impulse responses and counterfactual behavior

Impulse response functions are a foundational tool for dynamic causal analysis [66]. They characterize how localized shocks propagate through a system over time, revealing both the direction and persistence of causal influence. Importantly, impulse responses expose temporal structure that is not visible in static summaries or aggregate performance metrics.

From a scientific perspective, interpretable impulse responses exhibit recognizable qualitative properties [39]: effects evolve smoothly, decay or persist in ways consistent with known mechanisms, and maintain sign consistency unless substantive reversals are expected [50]. When impulse responses display erratic oscillations, abrupt sign changes, or sensitivity to minor perturbations, their scientific value is limited [50]. Evaluating impulse response behavior therefore provides a direct test of whether a model encodes plausible mechanisms rather than merely fitting observed trajectories.

Closely related are counterfactual trajectories, which trace system evolution under hypothetical interventions or shocks [34]. Whereas impulse responses focus on marginal effects, counterfactual analysis captures how perturbations propagate through the full system, potentially affecting many variables simultaneously [56]. For scientific reasoning, counterfactual trajectories should remain stable, bounded, and interpretable over time [73], reflecting whether effects amplify, dissipate, or reconfigure [74]. Counterfactual paths that exhibit implausible volatility or collapse indicate unreliable causal structure [29]. Because counterfactual behavior is not directly constrained by observed outcomes, it constitutes a stringent test of a model's internal coherence and causal plausibility.

3.3 Stability as causal evidence

A further dimension of scientific evidence concerns stability [69]. Causal conclusions should not hinge on idiosyncratic features of a particular sample, nor change dramatically under modest perturbations of the data or model specification. In dynamic settings, instability may appear as large fluctuations in inferred effects across time, regimes, or small structural variations.

Stability is therefore central to causal credibility [6]. Models that produce qualitatively consistent impulse responses and counterfactual trajectories across reasonable perturbations provide stronger evidence of underlying mechanisms than models whose behavior is highly sensitive. This notion of stability is behavioral rather than parametric and is distinct from classical statistical significance, which concerns sampling variability rather than robustness of dynamic behavior [32].

3.4 Implications for model evaluation

All these considerations suggest that evaluating dynamic causal models requires a broader evidentiary framework than prediction alone. Scientifically useful models should support coherent impulse responses, interpretable counterfactual trajectories, and stable behavior under perturbation, while maintaining reasonable predictive performance [41]. No single metric captures these properties; instead, they must be assessed through targeted behavioral diagnostics that probe how models respond to interventions and structural variation.

This perspective motivates the framework introduced in the following sections. Rather than treating prediction accuracy as the primary objective, we evaluate dynamic causal models based on the quality and stability of the causal behaviors they induce, allowing models to be assessed as instruments for scientific understanding rather than solely as forecasting tools.

4 DCNAR: Learned Networks as Structural Priors for Dynamic Causal Modeling

4.1 Dynamic causality with unknown structure

Let

$$\mathbf{x}_t = (x_{1,t}, \dots, x_{N,t})^\top, \quad t = 1, \dots, T, \quad (1)$$

denote an N -dimensional multivariate time series [22]. Dynamic causal modeling aims to characterize how directed dependencies among variables evolve over time, rather than assuming static interactions.

A general time-varying autoregressive representation [9] can be written as

$$\mathbf{x}_t = \sum_{\ell=1}^L \mathbf{A}_\ell(t) \mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_t, \quad (2)$$

where $\mathbf{A}_\ell(t) \in \mathbb{R}^{N \times N}$ encodes directed causal influence at lag ℓ and time t . Without additional constraints, estimating (2) is ill-posed in finite samples, particularly in short, heterogeneous, or nonstationary systems [37].

Most existing dynamic causal models address this by assuming that the support of $\mathbf{A}_\ell(t)$ is known in advance via a fixed adjacency matrix [26]

$$\mathbf{G} \in \{0, 1\}^{N \times N}, \quad (3)$$

such that

$$A_{\ell,ij}(t) = 0 \quad \text{whenever} \quad G_{ij} = 0. \quad (4)$$

This assumption presumes that the causal network is known, fixed, and correct [60]. In many scientific domains, however, causal structure is uncertain and itself a target of inquiry. DCNAR addresses this gap by replacing assumed structure with a learned and testable structural prior.

4.2 Overview of the DCNAR framework

Dynamic Causal Network Autoregression (DCNAR) is a two-stage framework that decouples causal structure discovery from dynamic causal estimation. Rather than jointly estimating a dense time-varying interaction tensor, DCNAR first infers a sparse directed network from data and then uses this network to constrain dynamic inference. This design allows causal influence to vary over time while maintaining interpretability and stability, and it enables impulse-response and counterfactual analysis under structural uncertainty.

4.3 Stage I: Neural autoregressive causal network discovery

In the first stage, DCNAR infers a candidate causal network using a neural additive autoregressive causal discovery model [11]. For each target variable $x_{i,t}$, the model takes the form

$$x_{i,t} = \sum_{j=1}^N \sum_{\ell=1}^L f_{ij\ell}(x_{j,t-\ell}) + \varepsilon_{i,t}, \quad (5)$$

where $f_{ij\ell}(\cdot)$ are univariate neural functions. The additive structure ensures that contributions from individual lagged variables are separable and interpretable.

Sparsity is induced through regularization on the collection $\{f_{ij\ell}\}$. Directed influence from variable j to i is summarized via an

aggregated causal score

$$S_{ij} = \sum_{\ell=1}^L \|f_{ij\ell}\|, \quad (6)$$

yielding a matrix $\mathbf{S} \in \mathbb{R}^{N \times N}$ of directed causal scores.

The causal score matrix does not represent regression coefficients or estimands with known sampling distributions. It is a sparse, directed Granger-causal network learned from multivariate time series, capturing predictive dependencies across variables rather than identified structural causal effects [11]. In DCNAR, it is treated as a *structural hypothesis* rather than as a final inferential object.

In practice, additional processing steps may be applied to $\mathbf{S}$, including stability assessment and necessity-based screening, to construct a sparse adjacency matrix. In the experiments reported here, we use a weighted adjacency matrix obtained via edge ablation, which retains only those directed relationships whose removal degrades out-of-sample performance, thereby prioritizing structurally necessary interactions. We emphasize that this edge-filtering procedure is an implementation tool for constructing the structural prior, not a claimed contribution of the present paper; the general framework for constructing and validating such restricted causal matrices is developed and benchmarked separately in [44], and its role in our pipeline is documented in the supplementary materials.

The defining methodological innovation of DCNAR is the interpretation of the learned network as a *structural prior*. Let

$$\widehat{\mathbf{G}} \in \{0, 1\}^{N \times N} \quad (7)$$

denote a directed adjacency matrix derived from the causal discovery stage. Rather than being assumed correct, $\widehat{\mathbf{G}}$ specifies which interactions are *admissible* in the dynamic model.

Formally, DCNAR constrains (2) by enforcing

$$A_{\ell,ij}(t) = 0 \quad \text{if} \quad \widehat{G}_{ij} = 0, \quad (8)$$

while allowing unrestricted time variation for admissible edges. This restriction reduces dimensionality, stabilizes estimation, and ensures that inferred dynamics correspond to explicit causal hypotheses derived from data. Because the structure is learned rather than imposed, its validity can be assessed indirectly through downstream dynamic behavior. In this sense, DCNAR treats causal structure as a falsifiable component of inference. In related work, we further develop this idea by imposing additional, empirically motivated restrictions on the learned causal matrix, such as stability- and necessity-based filtering, to evaluate which edges are substantively reliable rather than merely predictive [44].

4.4 Alternative Ways to Provide Structure for Dynamic Causal Modeling

Dynamic causal models such as time-varying network autoregressions require structural constraints to be identifiable and interpretable. When the underlying causal network is unknown, however, there are multiple plausible ways to supply such structure. In this section, we formalize and contrast alternative strategies for providing structure to dynamic causal models, and clarify how DCNAR differs from, and improves upon, these approaches. Importantly, the goal of this comparison is not to benchmark predictive

models, but to evaluate which sources of structure are scientifically defensible foundations for dynamic causal inference.

4.4.1 Coefficient-Based Structure from Linear VAR Models. A common approach to inferring structure in multivariate time series is to estimate a linear vector autoregressive (VAR) model with regularization [4]. In its general form, a VAR(L) model is written as

$$\mathbf{x}_t = \sum_{\ell=1}^L \mathbf{B}_\ell \mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_t, \quad (9)$$

where $\mathbf{B}_\ell \in \mathbb{R}^{N \times N}$ are lag-specific coefficient matrices.

In practice, regularization is used to stabilize estimation in high-dimensional or short-sample settings. Ridge VAR applies ℓ_2 regularization to shrink coefficient magnitudes uniformly, while sparse VAR variants employ ℓ_1 or elastic net penalties to encourage sparsity. Structural information is then often derived by thresholding estimated coefficients, yielding an adjacency matrix of the form

$$G_{ij}^{\text{VAR}} = \mathbb{I} \left(\sum_{\ell=1}^L |B_{\ell,ij}| > \tau \right), \quad (10)$$

where τ is a fixed threshold.

Time-varying VAR (TV-VAR) models extend this framework by allowing coefficients to evolve smoothly over time,

$$\mathbf{x}_t = \mathbf{B}(t) \mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t, \quad (11)$$

typically estimated using kernel smoothing or state-space formulations. In these models, dynamic dependence is captured through time-varying coefficients, but structural interpretation still relies on the magnitudes of $\mathbf{B}(t)$ or their time averages.

While coefficient-based VAR models offer simplicity and interpretability, they exhibit important limitations as sources of structure for dynamic causal modeling. First, even in TV-VAR formulations, structural interpretation remains tied to coefficient magnitude, which conflates direct and indirect effects and is sensitive to scaling and regularization. Second, in short or collinear time series, coefficient estimates, whether static or time-varying, can be unstable, leading to dense or highly variable inferred networks [63]. Finally, linear VAR-based structure reflects average linear dependence rather than causal influence in nonlinear or adaptive systems. As a result, although Ridge VAR and TV-VAR provide useful forecasting baselines and contextual comparators, coefficient-derived structure is poorly suited to serve as a stable structural prior for time-varying causal inference in complex systems.

4.4.2 Implicit Structure in Black-Box Predictive Models. An alternative strategy is to abandon explicit network structure altogether and rely on flexible black-box models, such as recurrent neural networks or long short-term memory (LSTM) architectures [46]. These models can be written abstractly as

$$\mathbf{x}_t = \mathcal{F}(\mathbf{x}_{t-1}, \dots, \mathbf{x}_{t-L}) + \boldsymbol{\varepsilon}_t, \quad (12)$$

where $\mathcal{F}(\cdot)$ is a high-capacity nonlinear function.

Such models often achieve strong predictive performance [31] and are effective at capturing complex temporal dependencies. However, they do not produce an explicit causal object corresponding to a directed network or time-varying causal influence [38]. As a result, they cannot support dynamic causal interpretation, structural validation, or hypothesis testing at the level of individual

relationships. From the perspective of dynamic causal modeling, implicit-structure models may therefore fail to provide the minimal representational requirements needed for scientific inference, regardless of their forecasting accuracy.

4.4.3 Static Assumed Networks. A third approach is to impose a fixed network structure derived from theory, prior studies, or expert knowledge [7]. In this case, the adjacency matrix $\mathbf{G}$ is specified exogenously and held constant throughout estimation.

While this strategy can be appropriate in domains with well-established causal mechanisms, it is problematic in many scientific contexts where causal relationships are uncertain, contested, or context-dependent [25]. Moreover, static assumed networks preclude the possibility of discovering previously unknown relationships and cannot adapt to structural change over time [15]. Consequently, this approach risks hard-coding potentially incorrect assumptions into dynamic models, undermining both interpretability and scientific validity.

4.4.4 Learned Explicit Structure via DCNAR. DCNAR occupies a distinct position among these alternatives by providing an explicit, data-driven causal structure that is neither assumed nor implicit. By learning a sparse directed network using NAVAR and treating it as a structural prior for time-varying modeling, DCNAR combines the interpretability of explicit networks with the flexibility of data-driven discovery.

Unlike coefficient-based approaches, the learned structure in DCNAR is nonlinear, lag-aware, and explicitly optimized for causal attribution rather than for linear approximation. Unlike black-box predictive models, DCNAR produces a concrete causal object that can be interrogated, ablated, and validated. Unlike assumed networks, the structure is inferred from data and subjected to empirical testing rather than imposed a priori.

This comparison clarifies that the central contribution of DCNAR is not a particular modeling choice, but a principled solution to the problem of supplying scientifically meaningful structure to dynamic causal models when such structure is unknown. By framing structure as a learned and testable prior, DCNAR enables dynamic causal inference in settings where existing approaches either assume away uncertainty or abandon causal interpretability altogether.

We deliberately do not adopt acyclicity-constrained or non-Gaussian causal discovery methods such as DYNOTEARS or time-lagged LiNGAM to construct $\hat{\mathbf{G}}$. These methods target graph recovery rather than dynamic causal inference, and their core assumptions are violated in this domain. Acyclicity is substantively inappropriate for democratic institutional systems, where feedback loops are the phenomenon of interest: legislative constraints shape executive behavior, which shapes electoral integrity, which in turn shapes legislative composition. LiNGAM’s non-Gaussianity assumption is also difficult to justify, since V-Dem indicators are expert-coded bounded ordinal measures aggregated to annual country-year observations, a process that yields approximately Gaussian marginals. Imposing either assumption would produce structure that contradicts established theory or violates the estimator’s premises, which is why we use a neural additive discovery stage that admits cyclic, nonlinear dependence.

4.5 Stage II: Time-varying network autoregression

Conditioned on the learned structural prior $\hat{\mathbf{G}}$, DCNAR estimates dynamic causal influence using a time-varying network autoregressive model. The baseline formulation follows the tvNAR framework [21], in which time variation enters through node-specific influence parameters while network structure is treated as fixed. This means time variation is captured at the level of each node’s overall influence rather than at the level of individual directed edges; we discuss the implications of this design choice in Section 7.

In the tvNAR(1) specification, the model is given by

$$\mathbf{x}_t = (\hat{\mathbf{G}} + \mathbf{I}) \Lambda(t) \mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t, \quad (13)$$

where $\hat{\mathbf{G}} \in \mathbb{R}^{N \times N}$ is the directed adjacency matrix supplied by the causal discovery stage, $\mathbf{I}$ is the identity matrix, and

$$\Lambda(t) = \text{diag}(\lambda_1(t), \dots, \lambda_N(t))$$

contains node-specific, smoothly varying influence parameters.

We extend the original tvNAR formulation by allowing for higher-order autoregressive dynamics. Specifically, DCNAR supports a tvNAR(p) model of the form

$$\mathbf{x}_t = \sum_{\ell=1}^p (\hat{\mathbf{G}} + \mathbf{I}) \Lambda_\ell(t) \mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_t, \quad (14)$$

where each $\Lambda_\ell(t)$ is a diagonal matrix of time-varying node influence parameters associated with lag ℓ . The tvNAR(1) model in Equation (13) is recovered as a special case when $p = 1$.

Time variation in $\Lambda_\ell(t)$ is estimated via kernel-based local smoothing over a normalized time index [21]. Crucially, the kernel is *one-sided*: the estimate at time t uses only observations at times $s \leq t$, so no future information contributes to any coefficient estimate. The smoothness of the resulting impulse responses therefore reflects genuine temporal regularization of past data rather than look-ahead interpolation. We verified that this behavior is not an artifact of the smoothing bandwidth: across bandwidths $h \in \{0.05, 0.10, 0.15, 0.20, 0.30, 0.40\}$, mean CRPS varies by less than 2×10^{-4} and impulse responses remain smooth, monotonic, and sign-consistent for every country examined (full results in the supplementary repository). In the experiments reported in this paper, we use the tvNAR(1) specification for clarity and stability; exploration of higher-order dynamics is deferred to supplementary analyses.

4.6 Impulse responses and counterfactual trajectories

The central methodological outputs of DCNAR are impulse-response functions (IRFs) and counterfactual trajectories derived from the time-varying, structurally constrained dynamics defined above. These objects form the primary basis for scientific interpretation and model comparison in this work.

Time-varying impulse response functions. Consider the tvNAR(p) model in Equation (14). Let

$$\mathbf{A}_\ell(t) = (\hat{\mathbf{G}} + \mathbf{I}) \Lambda_\ell(t), \quad (15)$$

and define the state-space representation

$$\mathbf{x}_t = \sum_{\ell=1}^p \mathbf{A}_\ell(t) \mathbf{x}_{t-\ell} + \boldsymbol{\varepsilon}_t. \quad (16)$$

For a given time index t_0 , the *local impulse response* of variable i to a unit shock in variable j at horizon h is defined recursively as

$$\Psi_{t_0}(0) = \mathbf{I}, \quad \Psi_{t_0}(h) = \sum_{\ell=1}^p \mathbf{A}_\ell(t_0 + h) \Psi_{t_0}(h - \ell), \quad h \geq 1, \quad (17)$$

with $\Psi_{t_0}(h) = \mathbf{0}$ for $h < 0$.

The (i, j) entry of $\Psi_{t_0}(h)$,

$$\text{IRF}_{i \leftarrow j}(t_0, h) = [\Psi_{t_0}(h)]_{ij}, \quad (18)$$

quantifies the effect at horizon h of a one-unit shock to variable j at time t_0 on variable i , conditional on the learned structural prior $\widehat{\mathbf{G}}$.

Because $\mathbf{A}_\ell(t)$ varies smoothly with time, impulse responses are themselves time-indexed objects, allowing causal propagation to differ across regimes. This distinguishes DCNAR from static VAR-based IRFs, which average causal behavior across the entire sample.

Structural role of learned networks. The learned adjacency matrix $\widehat{\mathbf{G}}$ enters the impulse-response computation through the support of $\mathbf{A}_\ell(t)$. If $\widehat{G}_{ij} = 0$, then $\text{IRF}_{i \leftarrow j}(t_0, h) = 0$ for all h unless mediated indirectly through other admissible paths. As a result, impulse responses reflect explicit causal hypotheses encoded by the learned structure, rather than dense or implicit interactions. This structural constraint is critical for interpretability. Without it, unconstrained time-varying models often produce oscillatory or unstable impulse responses that are difficult to reconcile with scientific theory.

Counterfactual trajectories. Impulse responses characterize marginal effects of localized shocks. To examine system-level behavior under sustained or composite interventions, we compute counterfactual trajectories.

Let $\mathbf{x}_t^{(0)}$ denote the observed trajectory generated by the fitted model, and let $\mathbf{x}_t^{(\delta)}$ denote the counterfactual trajectory under an intervention $\boldsymbol{\delta}_t$. The counterfactual dynamics are defined by

$$\mathbf{x}_t^{(\delta)} = \sum_{\ell=1}^p \mathbf{A}_\ell(t) \mathbf{x}_{t-\ell}^{(\delta)} + \boldsymbol{\delta}_t, \quad (19)$$

with $\boldsymbol{\delta}_t$ encoding the magnitude, timing, and target of the intervention.

System-level response measures. To summarize system-wide deviation under counterfactual scenarios, we define the aggregated response magnitude

$$\mathcal{R}(t) = \left\| \mathbf{x}_t^{(\delta)} - \mathbf{x}_t^{(0)} \right\|_2, \quad (20)$$

which captures the overall divergence between factual and counterfactual system trajectories.

This quantity allows direct comparison of dynamic behavior across models and across intervention types. In particular, models that produce unstable or incoherent dynamics tend to exhibit erratic or explosive $\mathcal{R}(t)$ paths, whereas DCNAR yields smooth, interpretable responses aligned with domain expectations.

Normalization and country-specific impulse responses. To facilitate comparison of counterfactual dynamics across countries with different baseline levels and scales, we additionally examine normalized system-level responses. Specifically, for each country c , we compute the L^2 normalization of the system state and normalize counterfactual deviations relative to the country-specific baseline trajectory. This normalization allows system-level responses to be interpreted as proportional deviations rather than absolute level changes, mitigating scale effects across panels.

In addition to system-level summaries, DCNAR supports impulse-response analysis at the level of individual variables within each country. For a fixed country c , impulse responses $\text{IRF}_{i \leftarrow j}^{(c)}(t_0, h)$ are computed using the country-specific time-varying coefficient paths $\mathbf{A}_\ell^{(c)}(t)$, allowing heterogeneous dynamic responses to be examined across institutional components and across countries. This enables direct inspection of how shocks to a specific variable propagate through the system in different national contexts, complementing the aggregated L^2 analysis shown in Figure 2.

Interpretation. Because impulse responses and counterfactual trajectories are computed conditional on a learned but explicit structural prior, their qualitative behavior provides evidence about the scientific plausibility of both the inferred structure and the dynamic model. In experiments, we use these objects, not forecast accuracy alone, as the primary basis for comparing DCNAR to alternative approaches.

5 Experiments

5.1 Data characteristics and empirical challenge

Our empirical evaluation is designed to reflect the conditions under which dynamic causal inference is most difficult in practice: many panels with short time series. The primary dataset consists of annual country-year observations from the Varieties of Democracy (V-Dem) project [17, 54], restricted to a modern window in which institutional indicators are most comparable across countries. The resulting panel contains a large number of countries (139) observed over a relatively short temporal horizon (35 years each).

This data regime poses a particular challenge for dynamic causal modeling. Short time series limit the feasibility of estimating richly parameterized time-varying models, while heterogeneity across panels makes it difficult to pool information without imposing strong structural assumptions. At the same time, the substantive questions motivating this study, such as how democratic components influence one another over time, and how shocks propagate through institutional systems, are inherently dynamic and causal in nature. The experimental setting therefore reflects a realistic and demanding use case rather than a favorable synthetic benchmark.

To assess robustness to temporal support, we additionally evaluate DCNAR on an extended panel from the same dataset with substantially longer time series but fewer countries (89 countries over 75 years). Results from this longer panel are reported in the supplementary materials. As discussed below, the qualitative behavior of DCNAR is consistent across the two settings, providing evidence that its dynamic causal conclusions are not an artifact of a particular panel configuration.

5.2 Predictive and distributional performance

We begin by comparing DCNAR to standard baselines using conventional predictive and distributional diagnostics, summarized in Appendix B with full detail in the supplementary materials, relative to Ridge VAR, TV-VAR, and LSTM (with Monte Carlo dropout). Figure 1 summarizes results across multi-horizon predictive distribution accuracy (CRPS), local one-step distributional accuracy, empirical coverage of nominal 90% prediction intervals, and a representative counterfactual impulse response.

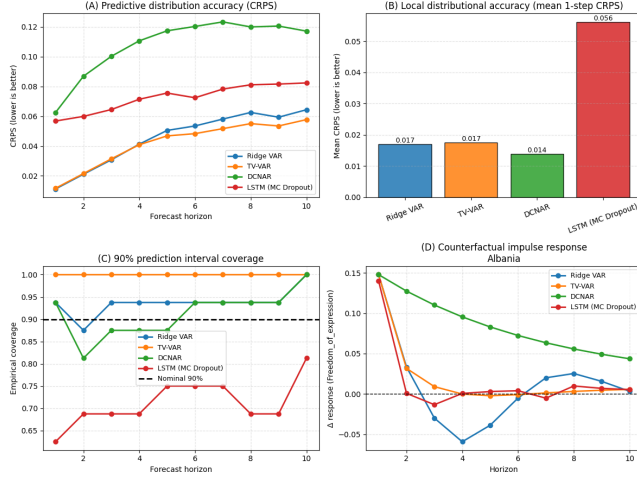


Figure 1: Comparison of DCNAR with Ridge VAR, TV-VAR, and LSTM (MC Dropout) across predictive (panel A, CRPS), distributional (panel B, local distributional accuracy - mean one-step-ahead CRPS), and causal diagnostics (panel C, nominal 90% prediction intervals across horizons) on the short-panel democracy dataset. Panel D shows representative counterfactual impulse response following a positive shock to freedom of expression (Albania).

Panels (A) and (B) show that DCNAR achieves predictive distribution accuracy comparable to Ridge VAR and TV-VAR across forecast horizons, and consistently outperforms the LSTM baseline. While DCNAR does not uniformly dominate all competitors on CRPS, differences across models are modest. Panel (C) further shows that DCNAR maintains stable, near-nominal coverage of 90% prediction intervals across horizons, indicating that its structural constraints do not compromise basic probabilistic calibration. By contrast, the LSTM baseline exhibits substantial undercoverage, reflecting miscalibrated uncertainty despite competitive point forecasts in some regimes.

These results establish that DCNAR satisfies a minimum credibility criterion: it is not meaningfully worse than established baselines on standard predictive and distributional metrics. This validation is necessary to ground subsequent causal analysis, but it is not the primary objective of the framework.

5.2.1 Impulse responses and counterfactual dynamics. The central contribution of DCNAR lies in the qualitative behavior of its impulse responses and counterfactual trajectories. Panel (D) of Figure

1 illustrates a representative counterfactual impulse response for an example country, Albania, following a positive shock to freedom of expression. DCNAR produces a smooth, monotonic decay pattern consistent with theoretical expectations, whereas TV-VAR and LSTM exhibit oscillations and sign reversals, and the Ridge response is erratic and difficult to interpret. Similar qualitative differences are observed across countries and variables.

Figure 2 extends this analysis to system-level counterfactual dynamics, summarized using the L^2 normalization across all democracy components for multiple countries. Solid lines show trajectories under a localized shock to freedom of expression, while dashed lines represent baseline evolution. The divergence between these trajectories captures the extent to which damage to a single democratic component propagates through the broader institutional system.

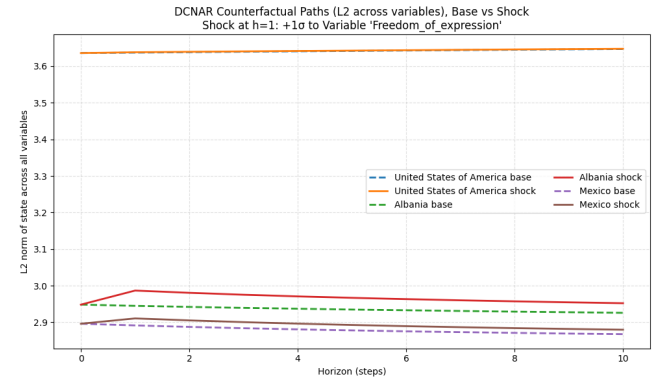


Figure 2: System-level counterfactual trajectories under DCNAR, summarized by the L^2 normalization across all democracy components, for Albania, the United States, and Mexico. Solid lines denote trajectories under a localized positive shock to freedom of expression at horizon $h = 1$; dashed lines denote corresponding baseline trajectories without intervention.

The resulting patterns reveal meaningful cross-country heterogeneity that aligns with existing political science theory [16]. Established democracies (such as the United States and the United Kingdom) exhibit relatively limited spillover effects, consistent with institutional resilience. Authoritarian regimes likewise display muted responses, as democratic components are already weak. In contrast, hybrid regimes and newer democracies (e.g., Albania and Mexico) show substantially stronger propagation of shocks, with cascading declines across democratic components. These patterns are intended as illustrative and hypothesis-generating, rather than definitive causal claims.

Importantly, DCNAR’s counterfactual trajectories remain smooth, bounded, and interpretable across cases. This behavior contrasts sharply with unconstrained or weakly constrained models, which often produce unstable or implausible system-level responses under the same perturbations. The coherence of DCNAR’s counterfactual dynamics indicates that the learned network operates effectively as a structural prior, constraining dynamic inference in a way that supports causal interpretation.

5.2.2 Stability across panel configurations. The qualitative patterns described above persist when DCNAR is applied to an extended panel with substantially longer time series. Despite differences in temporal support and sample composition, impulse responses and system-level counterfactual trajectories remain smooth, sign-consistent, and bounded. This robustness supports the interpretation that DCNAR captures persistent features of the underlying causal system rather than exploiting idiosyncrasies of a particular dataset. Results for the extended panel are reported in the supplementary materials.

5.3 Architectural ablation

To isolate the contribution of each DCNAR component, we conduct an ablation that holds all other components fixed and varies one at a time: replacing NAVAR’s neural discovery with a Ridge VAR–derived adjacency at matched density, replacing the time-varying $\Lambda(t)$ with its static (time-averaged) counterpart, and removing necessity-based sparsity (dense $\hat{G}$). Table 1 reports predictive accuracy (CRPS) alongside qualitative impulse-response diagnostics. Full per-country IRF and CRPS plots are provided in the supplementary repository.

Table 1: Architectural ablation on the short panel. Replacing neural discovery with linear VAR–derived structure degrades CRPS by 14.5%; all variants preserve sign-consistent, monotonic impulse responses.

Condition	Edges	CRPS	Sign cons. / Monotonic
DCNAR (full pipeline)	166	0.01311	3/3 / 3/3
VAR-derived $\hat{G}$ (no neural disc.)	166	0.01502	3/3 / 3/3
Static Λ (no time-variation)	166	0.01306	3/3 / 3/3
Dense $\hat{G}$ (no sparsity)	240	0.01323	3/3 / 3/3

The neural discovery stage is the component with the clearest predictive contribution: substituting a linear VAR–derived structure degrades CRPS by 14.5% and produces IRFs that systematically overestimate shock persistence. Removing time variation leaves one-step CRPS essentially unchanged (indeed marginally lower), which is expected in a highly persistent panel where short-horizon prediction is insensitive to smooth coefficient drift; the contribution of $\Lambda(t)$ is visible instead in longer-horizon causal behavior rather than in one-step accuracy. Removing sparsity modestly worsens CRPS, consistent with necessity-based filtering removing redundant propagation paths. Because $\hat{G}$ and $\Lambda(t)$ enter jointly through their product, these results should be read as sensitivity analyses rather than fully identified component effects.

5.4 Summary of experimental findings

The experimental results support two main conclusions. First, DCNAR meets standard predictive and distributional benchmarks, ensuring that its causal outputs are grounded in models with reasonable empirical fit. Second, and more importantly, DCNAR enables forms of dynamic causal analysis that are not supported by existing approaches. Its impulse responses and counterfactual trajectories are stable, interpretable, and theoretically meaningful,

even in short-panel settings where dynamic causal inference is typically unreliable. These findings reinforce the central claim of this study: in dynamic causal modeling, scientific value is determined not by marginal improvements in predictive accuracy, but by the coherence, stability, and interpretability of causal behavior under interrogation. DCNAR is explicitly designed to satisfy this criterion.

6 DCNAR and the Data-Generating Process

From the perspective of social sciences and economics, DCNAR enables analysis of human systems as complex adaptive processes in which institutional components respond endogenously to shocks and policy-relevant interventions propagate dynamically rather than instantaneously. Its primary contribution lies not in marginal improvements in predictive accuracy, but in the forms of scientific reasoning it enables. By producing stable and interpretable impulse responses and counterfactual trajectories under structural uncertainty, DCNAR provides direct access to features of the underlying data-generating process that are difficult to examine with existing approaches.

6.1 Persistence and transience of institutional shocks

Across countries and democracy components, DCNAR distinguishes between shocks with persistent effects and those that dissipate rapidly. Impulse responses typically exhibit an immediate impact followed by gradual decay, consistent with mechanisms of institutional adjustment rather than explosive feedback or instantaneous reversion. This behavior aligns with theoretical accounts of democratic change that emphasize inertia and gradual adaptation.

Importantly, these distinctions emerge from the qualitative form of impulse responses rather than from coefficient magnitudes or static correlations. Models that produce oscillatory or unstable responses obscure persistence and transience, limiting their usefulness for substantive interpretation.

6.1.1 Structural asymmetries across countries. System-level counterfactual trajectories reveal meaningful cross-country heterogeneity. While the overall pattern of shock propagation is similar, where smooth divergence is followed by stabilization, the magnitude and duration of responses vary across cases. Because DCNAR conditions dynamic inference on a learned but explicit causal structure, such heterogeneity can be interpreted as reflecting differences in institutional configuration and historical context rather than noise or model instability. In contrast, structure-free or black-box models provide little basis for attributing cross-country variation in dynamic behavior to substantive institutional features.

6.1.2 Recovered coefficient-path dynamics. A further form of scientific evidence comes from the estimated time-varying node-influence paths $\lambda_i(t)$ themselves. Without any domain supervision, DCNAR recovers regime dynamics that align with documented political history: for Albania, influence parameters fall sharply over 1991–1993 (post-communist institutional collapse) before stabilizing; for the United States they remain near-constant (institutional stability); and for Mexico they show a modest shift around 2000, coinciding with the end of single-party rule. By comparison, the corresponding

TV-VAR diagonal coefficient paths reach implausibly large magnitudes for Mexico and Albania that are difficult to reconcile with domain knowledge, and effective-network snapshots confirm that DCNAR’s structure remains stable across the observation window. These coefficient-path and effective-network visualizations are provided in full in the supplementary repository. This kind of substantive, historically interpretable read-out is inaccessible to static causal-discovery methods or black-box predictive models, and illustrates DCNAR’s intended role as a scientific instrument rather than a forecasting tool.

6.1.3 Stability of inferred mechanisms. A central finding is the stability of DCNAR’s dynamic causal behavior across panel configurations: despite substantial differences in temporal support between the 35-year and 75-year panels, impulse responses and counterfactual trajectories remain qualitatively consistent, suggesting that DCNAR captures persistent features of the underlying causal system rather than idiosyncrasies of a particular observation window. Such stability is essential for scientific use: models whose causal conclusions shift markedly under modest changes in temporal support offer limited epistemic value, whereas DCNAR’s coherence across data regimes supports its use as an instrument for investigating mechanisms rather than merely summarizing observed patterns.

7 Limitations and Ethical Considerations

While DCNAR expands the scope of dynamic causal analysis under structural uncertainty, its conclusions should be interpreted within clear bounds. First, DCNAR depends on the quality of the causal discovery stage. Although the framework treats learned structure as a testable prior rather than as ground truth, poor or unstable discovery can still constrain downstream inference. Ongoing work addresses this issue through stability and necessity diagnostics, but discovery quality remains a fundamental input. For example, we do not claim that the estimates the model currently displays are fully meaningful. They display the results of a DCNAR model run on 16 democracy indicators from the V-Dem database and no exogenous variables. Future work exploring complex democratic systems could focus on analyzing a much larger dataset using the DCNAR approach.

Second, finite-sample limitations remain important, particularly in short-panel settings. While the structural prior stabilizes dynamic estimation, time-varying causal inference is inherently data-hungry, and very short or noisy series may still limit resolution of fine-grained temporal dynamics.

Third, DCNAR does not claim to recover true structural causality in the sense of fully identified structural causal models. The causal relationships inferred here are Granger-causal, empirical and behavioral, grounded in predictive dependence and dynamic response rather than in controlled intervention. The framework is intended to support scientific exploration and hypothesis evaluation, not definitive causal identification.

Fourth, time variation in DCNAR enters through node-specific influence parameters $\Lambda(t)$ rather than edge-specific ones. The model therefore captures how the overall influence of each variable evolves over time, but not how the directed relationship between a specific pair of variables changes. This is a deliberate identifiability choice:

fully edge-level time variation would require $O(N^2)$ time-varying parameters rather than $O(N)$, which is ill-posed given the short per-unit series in panel democracy data. Relaxing this constraint through low-rank time-varying interaction structure is a natural extension and is developed in separate work.

These limitations reflect deliberate design choices. DCNAR prioritizes interpretability, stability, and falsifiability over strong identification claims, aligning it with the norms of observational scientific research in complex systems.

Ethical use of data and informed consent. This study uses the Varieties of Democracy (V-Dem) dataset, which is publicly available, aggregated at the country–year level, and contains no personally identifiable information. No human subjects were involved at any stage of the research, and informed-consent requirements therefore do not apply. Beyond data provenance, three considerations merit explicit statement given the politically sensitive domain. First, the analysis is observational and non-interventional: we make no prescriptive policy claims, and our results characterize dynamic dependence in historical data rather than recommend actions. Second, as noted above, all inferred relationships are Granger-causal rather than structurally identified; causal language throughout should be read in this restricted, behavioral sense. Third, because causal claims about political institutions are susceptible to misuse, we emphasize that DCNAR’s outputs are hypothesis-generating instruments for scientific inquiry, not definitive findings about any country’s institutions. To support scrutiny and reproducibility, all data and code are openly available (Appendix A).

8 Conclusion

This paper introduced DCNAR as a framework for dynamic causal analysis when causal structure is unknown. By integrating causal discovery with time-varying network autoregression through learned structural priors, DCNAR enables forms of reasoning that are largely inaccessible to existing models.

Our results show that DCNAR matches standard baselines on predictive and distributional metrics while delivering qualitatively different, and scientifically more useful, causal behavior. Its impulse responses and counterfactual trajectories are stable, interpretable, and consistent with theoretical expectations, even in short-panel settings where dynamic causal inference is typically unreliable.

The broader implication is methodological. In scientific applications, the value of AI models lies not only in prediction, but in their ability to behave as instruments for understanding—to expose mechanisms, test hypotheses, and support counterfactual reasoning. DCNAR reframes dynamic causal modeling around interpretable behavior under perturbation rather than accuracy alone, arguing that such models should be judged by the coherence and stability of the causal stories they enable. We hope it encourages further development of AI systems designed explicitly for scientific inquiry.

Acknowledgments

This work was supported by funding from three research institutes at the University of Notre Dame: the Kellogg Institute for International Studies, the Franco Family Institute for Liberal Arts and the Public Good, and the Lucy Family Institute for Data & Society.

Generative AI Disclosure. Generative AI software (ChatGPT) was used to edit and improve the quality of author-written text.

References

- [1] Sarah F. Ackley, Justin Lessler, and M. Maria Glymour. 2022. Dynamical modeling as a tool for inferring causation. *American Journal of Epidemiology* 191, 1 (2022), 1–6.
- [2] Ulf Andersson, Alvaro Cuervo-Cazurra, and Bo Bernhard Nielsen. 2019. Explaining interaction effects within and across levels of analysis. In *Research Methods in International Business*. Springer International Publishing, Cham, 331–349.
- [3] Mirko Armillotta and Konstantinos Fokianos. 2023. Nonlinear network autoregression. *The Annals of Statistics* 51, 6 (2023), 2526–2552. doi:10.1214/23-AOS2345
- [4] Giovanni Ballarin. 2025. Ridge regularized estimation of VAR models for inference. *Journal of Time Series Analysis* 46, 2 (2025), 235–257.
- [5] Manal Benhamza, Marianne Clausel, and Myriam Tami. 2025. Counterfactual Robustness: A Framework to Analyze the Robustness of Causal Generative Models Across Interventions. In *Machine Learning and Knowledge Discovery in Databases: Research Track (ECML PKDD 2025)*. Springer Nature Switzerland, Cham, 391–408. doi:10.1007/978-3-032-05962-8_23
- [6] Andreas Bergh and Philipp C. Wichardt. 2024. *On credibility and causality in economics: A critical appraisal*. Technical Report. IFN Working Paper.
- [7] Matthew Blackwell. 2013. A framework for dynamic causal inference in political science. *American Journal of Political Science* 57, 2 (2013), 504–520.
- [8] Stephan Bongers, Tineke Blom, and Joris M. Mooij. 2018. Causal modeling of dynamical systems. arXiv:1803.08784 [cs.LG]
- [9] Laura F. Bringmann, Ellen L. Hamaker, Daniel E. Vigo, André Aubert, Denny Borsboom, and Francis Tuerlinckx. 2017. Changing dynamics: Time-varying autoregressive models using generalized additive modeling. *Psychological Methods* 22, 3 (2017), 409–425.
- [10] Marc J. Buehner. 2012. Understanding the past, predicting the future: Causation, not intentional action, is the root of temporal binding. *Psychological science* 23, 12 (2012), 1490–1497.
- [11] Bart Bussmann, Jannes Nys, and Steven Latré. 2021. Neural additive vector autoregression models for causal discovery in time series. In *International Conference on Discovery Science*. Springer International Publishing, Cham, 446–460.
- [12] Guglielmo Maria Caporale and Nikitas Pittis. 1997. Causality and forecasting in incomplete systems. *Journal of Forecasting* 16, 6 (1997), 425–437.
- [13] Alexandra Cirone and Thomas B. Pepinsky. 2022. Historical persistence. *Annual Review of Political Science* 25, 1 (2022), 241–259.
- [14] Philip J. Clare, Timothy A. Dobbins, and Richard P. Mattick. 2019. Causal models adjusting for time-varying confounding—a systematic review of the literature. *International Journal of Epidemiology* 48, 1 (2019), 254–265.
- [15] Gregory F. Cooper. 1999. An overview of the representation and discovery of causal relationships using Bayesian networks. *Computation, Causation, and Discovery* (1999), 4–62.
- [16] Michael Coppedge, Amanda B. Edgell, Carl Henrik Knutsen, and Staffan I. Lindberg. 2022. *Why democracies develop and decline*. Cambridge University Press, Cambridge.
- [17] Michael Coppedge, John Gerring, Carl Henrik Knutsen, Staffan I. Lindberg, Jan Teorell, et al. 2025. V-Dem Country-Year Dataset v15. Varieties of Democracy (V-Dem) Project. doi:10.23696/vdemds25
- [18] Skyler J. Cranmer and Bruce A. Desmarais. 2017. What can we learn from predictive modeling? *Political Analysis* 25, 2 (2017), 145–166.
- [19] Petros T. Damos. 2024. On formal limitations of causal ecological networks. *Philosophical Transactions of the Royal Society B* 379, 1909 (2024), 20230170.
- [20] Annamaria Defilippo, Federico Manuel Giorgi, Pierangelo Veltri, and Pietro Hiram Guzzi. 2024. Understanding complex systems through differential causal networks. *Scientific Reports* 14, 1 (2024), 27431.
- [21] Yi Ding, Xuening Zhu, Rui Pan, and Bo Zhang. 2025. Network Vector Autoregression with Time-Varying Nodal Influence. *Computational Economics* (2025), 1–27.
- [22] Magnus P. Ekstrom and Thomas L. Marzetta. 1981. Fundamentals of multidimensional time-series analysis. In *Identification of Seismic Sources—Earthquake or Underground Explosion*. NATO Advanced Study Institutes Series, Vol. 74. Springer Netherlands, Dordrecht, 615–647. doi:10.1007/978-94-009-8531-5_35
- [23] Uwe Engel. 2021. *Causal and predictive modeling in computational social science*. Taylor and Francis.
- [24] Muhammad Ashraf Faheem, Muhammad Aslam, and Sridevi Kakolu. 2024. *Enhancing financial forecasting accuracy through AI-driven predictive analytics models*. doi:10.13140/RG.2.2.36214.20800
- [25] Tulia G. Falsetti and Julia F. Lynch. 2009. Context and causal mechanisms in political analysis. *Comparative Political Studies* 42, 9 (2009), 1143–1166.
- [26] Karl J. Friston, Lee Harrison, and Will Penny. 2003. Dynamic causal modelling. *Neuroimage* 19, 4 (2003), 1273–1302.
- [27] Karl J. Friston, Katrin H. Preller, Chris Mathys, Hayriye Cagnan, Jakob Heinze, Adeel Razi, and Peter Zeidman. 2019. Dynamic causal modelling revisited. *Neuroimage* 199 (2019), 730–744.
- [28] John Gerring. 2005. Causation: A unified framework for the social sciences. *Journal of Theoretical Politics* 17, 2 (2005), 163–198.
- [29] Tobias Gerstenberg and Simon Stephan. 2021. A counterfactual simulation model of causation by omission. *Cognition* 216 (2021), 104842.
- [30] Clark Glymour, Kun Zhang, and Peter Spirtes. 2019. Review of causal discovery methods based on graphical models. *Frontiers in Genetics* 10 (2019), 524.
- [31] Riccardo Guidotti, Anna Monreale, Salvatore Ruggieri, Franco Turini, Fosca Giannotti, and Dino Pedreschi. 2018. A survey of methods for explaining black box models. *Comput. Surveys* 51, 5 (2018), 1–42.
- [32] Joseph Y. Halpern. 2016. Appropriate causal models and the stability of causation. *The Review of Symbolic Logic* 9, 1 (2016), 76–102.
- [33] Jonas M. B. Haslbeck, Laura F. Bringmann, and Lourens J. Waldorp. 2021. A tutorial on estimating time-varying vector autoregressive models. *Multivariate Behavioral Research* 56, 1 (2021), 120–149.
- [34] Benjamin C. Herd and Simon Miles. 2019. Detecting causal relationships in simulation models using intervention-based counterfactual analysis. *ACM Transactions on Intelligent Systems and Technology* 10, 5 (2019), 1–25. doi:10.1145/3322123
- [35] Jake M. Hofman, Amit Sharma, and Duncan J. Watts. 2017. Prediction and explanation in social systems. *Science* 355, 6324 (2017), 486–488.
- [36] Jake M. Hofman, Duncan J. Watts, Susan Athey, Filiz Garip, Thomas L. Griffiths, Jon Kleinberg, Helen Margetts, Sendhil Mullainathan, Matthew J. Salganik, Simine Vazire, Alessandro Vespignani, and Tal Yarkoni. 2021. Integrating explanation and prediction in computational social science. *Nature* 595, 7866 (2021), 181–188.
- [37] Richard Horský. 2017. Non-stationary Stochastic Sequences as Solutions to Ill-posed Problems. Working paper.
- [38] Tsung-Yu Hsieh. 2021. *Explainable Predictive Modeling and Causal Effect Estimation from Complex Time-Varying Data*. Ph.D. Dissertation. The Pennsylvania State University.
- [39] Atsushi Inoue and Lutz Kilian. 2013. Inference on impulse response functions in structural VAR models. *Journal of Econometrics* 177, 1 (2013), 1–13.
- [40] Maksims Ivanovs, Roberts Kadikis, and Kaspars Ozols. 2021. Perturbation-based methods for explaining deep neural networks: A survey. *Pattern Recognition Letters* 150 (2021), 228–234.
- [41] Benjamin James. 2024. *Achieving Coherence: Modeling Complexity in Dynamic Systems*. Kindle Direct Publishing.
- [42] Yeliz Karaca and Dumitru Baleanu. 2022. Evolutionary mathematical science, fractional modeling and artificial intelligence of nonlinear dynamics in complex systems. *Chaos Theory and Applications* 4, 3 (2022), 111–118.
- [43] Milja Kurki. 2008. *Causation in international relations: Reclaiming causal analysis*. Cambridge University Press, Cambridge, UK.
- [44] Valentina V. Kuskova, Dmitry Zaytsev, and Michael Coppedge. 2026. Beyond Coefficients: Forecast-Necessity Testing for Interpretable Causal Discovery in Nonlinear Time-Series Models. In *Proceedings of the 39th International Florida Artificial Intelligence Research Society Conference (FLAIRS)*. doi:10.32473/flairs.39.1.141791 arXiv:2604.18751.
- [45] Valentina V. Kuskova, Dmitry Zaytsev, and Michael Coppedge. 2026. Function-Valued Causal Influence in Nonlinear Time Series. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. arXiv:2605.26408.
- [46] Benjamin Lindemann, Timo Müller, Hannes Vietz, Nasser Jazdi, and Michael Weyrich. 2021. A survey on long short-term memory networks for time series prediction. *Procedia CIRP* 99 (2021), 650–655.
- [47] Helmut Lütkepohl. 2018. Impulse response function. In *The New Palgrave Dictionary of Economics*. Palgrave Macmillan, London, 6141–6145.
- [48] Ricardo P. Masini, Marcelo C. Medeiros, and Eduardo F. Mendes. 2023. Machine learning advances for time series forecasting. *Journal of Economic Surveys* 37, 1 (2023), 76–111.
- [49] Christoph Molnar and Timo Freiesleben. 2024. *Supervised Machine Learning for Science: How to stop worrying and love your black box*. Independently published.
- [50] Angela Montoya, Ed M. Habtour, and Fernando Moreu. 2020. Quantifying Information without Entropy: Identifying Intermittent Disturbances in Dynamical Systems. *Entropy* 22, 11 (2020), 1199. doi:10.3390/e22111199
- [51] Stephen L. Morgan and Christopher Winship. 2014. *Counterfactuals and causal inference: Methods and principles for social research*. Cambridge University Press, Cambridge, UK.
- [52] Steve Morlidge. 2013. How good is a “good” forecast? Forecast errors and their avoidability. *Foresight: The International Journal of Applied Forecasting* 30 (2013), 5–11.
- [53] Mohamed N. Nounou. 2006. Multiscale finite impulse response modeling. *Engineering Applications of Artificial Intelligence* 19, 3 (2006), 289–304.
- [54] Daniel Pemstein, Kyle L. Marquardt, Eitan Tzelgov, Yi-ting Wang, Juraj Medzihorsky, Joshua Krusell, Farhad Miri, and Johannes von Römer. 2025. *The V-Dem Measurement Model: Latent Variable Analysis for Cross-National and Cross-Temporal Expert-Coded Data*. V-Dem Working Paper 21, 10th ed. University of Gothenburg, Varieties of Democracy Institute.
- [55] Simon M. Potter. 2000. Nonlinear impulse response functions. *Journal of Economic Dynamics and Control* 24, 10 (2000), 1425–1446.

- [56] Mattia Proserpi, Yi Guo, Matt Sperrin, James S. Koopman, Jae S. Min, Xing He, Shannan Rich, Mo Wang, Iain E. Buchan, and Jiang Bian. 2020. Causal inference and counterfactual prediction in machine learning for actionable healthcare. *Nature Machine Intelligence* 2, 7 (2020), 369–375.
- [57] Atul Rawal, Adrienne Raglin, Danda B. Rawat, Brian M. Sadler, and James McCoy. 2025. Causality for trustworthy artificial intelligence: status, challenges and perspectives. *Comput. Surveys* 57, 6 (2025), 1–30. doi:10.1145/3665494
- [58] Lauren N. Ross. 2024. What is social structural explanation? A causal account. *Noûs* 58, 1 (2024), 163–179.
- [59] Barbara Rossi. 2024. Recent developments in forecast evaluation. In *Handbook of Research Methods and Applications in Macroeconomic Forecasting*. Edward Elgar Publishing, 274.
- [60] Benjamin M. Rottman and Reid Hastie. 2014. Reasoning about causal relationships: Inferences on causal networks. *Psychological Bulletin* 140, 1 (2014), 109–139.
- [61] Cynthia Rudin, Chaofan Chen, Zhi Chen, Haiyang Huang, Lesia Semenova, and Chudi Zhong. 2022. Interpretable machine learning: Fundamental principles and 10 grand challenges. *Statistics Surveys* 16 (2022), 1–85.
- [62] Francisco Ruge-Murcia. 2025. Using generalized impulse response functions to estimate nonlinear dynamic models. *Econometric Reviews* (2025), 1–24.
- [63] Román Salmerón Gómez, José García Pérez, María del Mar López Martín, and Catalina B. García García. 2016. Collinearity diagnostic applied in ridge estimation through the variance inflation factor. *Journal of Applied Statistics* 43, 10 (2016), 1831–1849. doi:10.1080/02664763.2015.1120712
- [64] Andrew R. Sedler, Christopher Versteeg, and Chethan Pandarinath. 2023. Expressive architectures enhance interpretability of dynamics-based neural population models. *Neurons, Behavior, Data Analysis, and Theory* 10 (2023), 1–22. doi:10.51628/001c.73987
- [65] Andrew Smart and Atoosa Kasirzadeh. 2025. Beyond model interpretability: Socio-structural explanations in machine learning. *AI & Society* 40, 4 (2025), 2045–2053.
- [66] Norman R. Swanson and Clive W. J. Granger. 1997. Impulse response functions based on a causal approach to residual orthogonalization in vector autoregressions. *J. Amer. Statist. Assoc.* 92, 437 (1997), 357–367.
- [67] Ai Hui Tan. 2022. Kernel-based impulse response estimation using perturbation signals with harmonic suppression. *IEEE Control Systems Letters* 7 (2022), 607–612.
- [68] Paul Thagard. 2012. *The cognitive science of science: Explanation, discovery, and conceptual change*. MIT Press, Boston, MA.
- [69] John Thornes. 2020. Problems in the identification of stability and structure from temporal data series. In *Space and Time in Geomorphology*. Routledge, 327–353.
- [70] Payal Varshney, Adriano Lucieri, Christoph Balada, Andreas Dengel, and Sheraz Ahmed. 2024. Generating counterfactual trajectories with latent diffusion models for concept discovery. In *International Conference on Pattern Recognition*. Springer Nature Switzerland, 138–153.
- [71] Andreas Wagner. 1999. Causality in complex systems. *Biology and Philosophy* 14 (1999), 83–101.
- [72] Erik Weber and Bert Leuridan. 2008. Counterfactual causality, empirical research, and the role of theory in the social sciences. *Historical Methods: A Journal of Quantitative and Interdisciplinary History* 41, 4 (2008), 197–201.
- [73] Juliane Weillbach, Sebastian Gerwinn, Karim Barsim, and Martin Fränzle. 2024. Counterfactual-Based Root Cause Analysis for Dynamical Systems. In *Machine Learning and Knowledge Discovery in Databases: Research Track (ECML PKDD 2024)*. Springer Nature Switzerland, Cham, 303–319. doi:10.1007/978-3-031-70365-2_18
- [74] Michael R. Wickens and Roberto Motto. 2001. Estimating shocks and impulse response functions. *Journal of Applied Econometrics* 16, 3 (2001), 371–387. doi:10.1002/jae.617
- [75] Xuening Zhu, Rui Pan, Guodong Li, Yuewen Liu, and Hansheng Wang. 2017. Network vector autoregression. *The Annals of Statistics* 45, 3 (2017), 1096–1123.

A Data and Code Availability

All data, code, and supplementary materials are available at <https://github.com/vkuskova/SIGKDD2026-463>. The repository contains the complete experimental pipeline, full implementation details for DCNAR and all baselines, the bandwidth-sensitivity and architectural-ablation results, and the time-varying coefficient-path and effective-network visualizations referenced in the main text. A companion supplementary materials document accompanies this paper and collects the extended methodological detail summarized below.

B Experimental Details

This appendix summarizes the essential specifications needed to interpret the results in Section 5. Full implementation detail, the edge-ablation construction of the structural prior, computational considerations, and the extended-panel analysis are provided in the supplementary materials and the repository above.

Baselines. The *Ridge VAR* baseline is a first-order vector autoregression $\mathbf{x}_t = \mathbf{B} \mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t$, with $\mathbf{B}$ estimated by ridge-regularized least squares and predictive uncertainty obtained by empirical residual resampling. The *TV-VAR* baseline replaces $\mathbf{B}$ with a smoothly time-varying $\mathbf{B}(t)$ estimated by kernel-weighted ridge regression using a Gaussian kernel of fixed bandwidth. The *LSTM* baseline is a single-layer recurrent network trained for one-step prediction, with predictive distributions obtained via Monte Carlo dropout.

Dynamic inference. The primary specification is tvNAR(1),

$$\mathbf{x}_t = (\hat{\mathbf{G}} + \mathbf{I}) \Lambda(t) \mathbf{x}_{t-1} + \boldsymbol{\varepsilon}_t, \quad (21)$$

where $\hat{\mathbf{G}}$ is the learned structural prior and $\Lambda(t)$ is a diagonal matrix of node-specific, smoothly varying influence parameters estimated by *one-sided* kernel smoothing over a normalized, country-aware time index (no future observations contribute to any estimate). The structural prior $\hat{\mathbf{G}}$ is obtained from the NAVAR causal score matrix via necessity-based edge ablation; the methodology is developed and benchmarked in [44] and documented in the supplementary materials.

Evaluation metrics. Predictive distribution accuracy is measured by the Continuous Ranked Probability Score (CRPS), a proper scoring rule,

$$\text{CRPS}(F_t, y_t) = \int_{-\infty}^{\infty} (F_t(z) - \mathbb{I}\{z \geq y_t\})^2 dz, \quad (22)$$

averaged across horizons, variables, countries, and forecast origins. Uncertainty calibration is assessed by empirical coverage of nominal $(1 - \alpha)$ prediction intervals $[Q_{\alpha/2}(F_t), Q_{1-\alpha/2}(F_t)]$; we report $\alpha = 0.10$ (nominal 90% intervals). Causal behavior is evaluated through counterfactual impulse responses $\text{IRF}_{i \leftarrow j}(t_0, h) = \mathbf{x}_{i,t_0+h}^{(\delta)} - \mathbf{x}_{i,t_0+h}^{(0)}$ and the system-level L^2 deviation $\mathcal{R}(t_0 + h) = \|\mathbf{x}_{t_0+h}^{(\delta)} - \mathbf{x}_{t_0+h}^{(0)}\|_2$.

Robustness and ablations. The qualitative findings are stable across a six-value bandwidth sweep ($h \in \{0.05, \dots, 0.40\}$; mean CRPS varies by $< 2 \times 10^{-4}$) and are supported by an architectural ablation isolating each DCNAR component (Table 1). The extended 89-country, 75-year panel reproduces the main qualitative behavior. Full tables, figures, and per-country plots for all three analyses are provided in the supplementary materials and the repository.